

Bo Shang

Field Data Engineer | Data Ops & Infrastructure
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Field-first data engineer with 5+ years hands-on experience deploying and scaling **real-world sensor data collection systems**. Expert in building robust, configuration-portable data pipelines from field sensors through cloud infrastructure. Proven ability to translate field constraints into scalable solutions: validated multi-modal data ingestion (LiDAR, RGB, thermal), quality assurance protocols, and production-grade infrastructure for autonomous systems and infrastructure monitoring.

EXPERIENCE

- **City University of New York — AI & Mobility Research Lab** New York, NY
Field Data Engineer — Autonomous Vehicles Jul 2025 – Present
 - Operationalized **fixed roadside LiDAR data collection** across multiple sensor sites; built **configuration-portable inference pipeline** to handle heterogeneous sensor variants (beam count, frame rate, mounting height)
 - Designed and validated data ingestion from production LiDAR sensors; implemented quality metrics for multi-frame point-cloud reconstruction and sensor reliability assessment
 - Established data labeling and annotation protocols for vulnerable road user classification; coordinated with field teams to validate sensor coverage and performance at 3+ sites
 - Built reproducible dataset versioning and format conversion infrastructure across 5+ sensor configuration variants
- **City University of New York — Robotics Lab** New York, NY
Field Data Engineer — Infrastructure Inspection Dec 2022 – Dec 2024
 - Owned end-to-end data pipeline for **robotic bridge inspection**: field data collection (drone/robot imagery + multi-sensor metadata), ingestion, storage, processing, and delivery at scale
 - Architected and deployed **WebODM-based data platform** integrating automated image preprocessing, 3D reconstruction, labeling workflows, and visualization; served 10+ field operations teams
 - Implemented data quality assurance protocols: image deduplication, anomaly detection, missing-data flagging; reduced bad-data insertion by 85%
 - Designed robust data collection workflows in GPS-denied and high-interference environments; validated field-to-cloud data integrity across 50+ sites
 - Pushed inference at scale to AWS (EC2, S3); optimized inference latency and cost for automated defect localization on 100k+ images
- **Missouri University of Science and Technology** Rolla, MO
Embedded Data Systems Engineer Jan 2020 – Nov 2022
 - Deployed real-time vision processing on **NVIDIA Jetson edge hardware** (Nano, TX2, Xavier) for field robotics; optimized inference pipelines for power and latency constraints
 - Built data quality evaluation framework for visual odometry systems; integrated motion-capture ground truth (VICON) for validation in GPS-denied environments
 - Designed and tested data collection protocols for autonomous flight in harsh conditions; characterized sensor reliability and failure modes
- **Vaughn College of Aeronautics and Technology** New York, NY
Technical Instructor 2023 – 2024
 - Taught applied courses on AI systems, research methodology, and robotics control; mentored 20+ students on practical deployment of autonomous systems
- **University of California, Merced** Merced, CA
Field Robotics Systems Engineer Sep 2015 – Sep 2017
 - Co-led development of **autonomous robotic platform for GPS-denied exploration**; integrated inertial + visual odometry, sensor fusion, and robust localization for cave deployment
 - Built custom embedded data-collection software on Linux UAVs; validated sensor data in real-world environments with high noise and degraded signals

KEY DATA OPERATIONS PROJECTS

- **Heterogeneous Sensor Data Ingestion (2024–present)**: Built configuration-portable pipelines handling LiDAR variants: 16/32/64-beam, 10–20 Hz frame rates, 3+ mounting configs. Validated end-to-end data consistency across all variants; reduced data-preparation time by 60%.
- **Bridge Inspection Data Platform (2022–present)**: Designed and operated cloud-based system for drone/robot imagery ingestion, preprocessing, 3D reconstruction, and interactive delivery. Handled 100k+ images; achieved 99.2% uptime; reduced analyst turnaround from 2 weeks to 2 days.
- **Data Quality & Validation Framework (2023–2024)**: Implemented automated quality checks for field data: image blur detection, sensor calibration validation, metadata consistency. Flagged 15–20% problematic data before downstream processing.
- **Edge Inference at Scale (2020–present)**: Optimized neural network inference for NVIDIA Jetson platforms under strict power and latency constraints. Deployed on 50+ field robots; achieved 15–50x speedup via quantization and model optimization.

CORE COMPETENCIES

- **Data Ops & Pipelines:** End-to-end sensor data ingestion, ROS / ROS 2, MCAP, Protobuf, ETL, data versioning, format conversion, quality assurance, metadata management
- **Cloud & Infrastructure:** AWS (S3, EC2, Lambda, SageMaker), Docker, Kubernetes, CI/CD (GitHub Actions), observability and monitoring
- **Edge Computing:** NVIDIA Jetson (Nano/TX2/Xavier), embedded Linux, real-time constraints, power-aware optimization, sensor driver development
- **Field Engineering:** Multi-modal sensor integration (LiDAR, RGB, thermal, IMU), sensor calibration and validation, GPS-denied localization, robustness under real-world constraints
- **Programming:** Python (primary), C++, MATLAB; familiar with Go, Rust for systems work
- **Data Formats & Validation:** JSON, CSV, HDF5, custom binary formats, schema validation, data format migrations

EDUCATION

- **Northeastern University & University of California, Merced** Shenyang, China / Merced, CA
Ph.D. in Pattern Recognition & Intelligent Systems *Aug 2013 – Dec 2020*
- **Northeastern University** Shenyang, China
M.Eng. in Pattern Recognition & Intelligent Systems; B.Eng. in Automation *Aug 2007 – Dec 2013*

PATENTS & RECOGNITION

- **US 12,296,994** (granted May 2025) — Unmanned vehicle system design and deployment
- **Best System Control & Best Mission Planning Awards** — International Aerial Robotics Competition (AUVSI), 2014